

Interpreting, correcting and completing algorithms

Revise 360

Worksheet for the 360° experience: Algorithm repair clinic

Name: _____

Class: _____

Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Algorithm repair clinic. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Interpret algorithms using sample inputs.
- Correct faulty conditions and updates.
- Complete missing algorithm steps.

Starter: before you put the headset on

A total-cost algorithm adds price and quantity. Is this the correct calculation?

.....

.....

Key terminology

Logic error

.....

.....

Refinement

.....

.....

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Read what is written

Turn right

Key facts in your own words:

Example to trace:

```
x = 4
y = x + 3
x = y * 2
print(x)
```

Challenge: What is printed and why?

x = 2; y = x * 5; print(y + 1). Output?

x = 4; y = x + 3. What is y?

2 Correct an operation

Turn right

Key facts in your own words:

Example to trace:

```
Need: total cost
quantity = 3; price = 5
Wrong: total = quantity + price
Correct: quantity * price
```

Challenge: What are the wrong and correct results?

A rectangle area uses length + width. Correction?

For length 6 and width 2, area is...

3 Correct a boundary

Turn around

Key facts in your own words:

Example to trace:

RULE: score at least 40

Wrong: score > 40

Correct: score >= 40

Test: 39, 40, 41

Challenge: Which test exposes the error?

The rule is fewer than 10. Which condition fits?

Which value exposes > 40 used for at least 40?

4 Complete an update

Turn around

Key facts in your own words:

Example to trace:

```
n = 3
```

```
while n > 0
```

```
    print(n)
```

```
    [missing update]
```

```
endwhile
```

Challenge: Complete the update so it prints 3, 2, 1.

Which update makes this countdown terminate?

A countdown from 3 using $n = n - 1$ should stop at...

5 Refine an omitted case

Turn left

Key facts in your own words:

Example to trace:

```
if a > b: A wins
else: B wins
```

Fault: `a == b`

Needs: draw result

Challenge: What happens incorrectly when `a == b`?

How should the two-player comparison be refined?

A draw should be reported when...

6 Recheck the correction

Turn left

Key facts in your own words:

Example to trace:

```
Fix: score >= 40
Retest: 40 -> pass
Check: 39 -> fail
Check: 41 -> pass
```

Challenge: Why rerun a previously successful case?

Order the repair process.

Put these steps in order: Retest failing and working cases / Correct the identified fault / Trace the faulty algorithm / State the expected behaviour

Retesting a previously working input checks...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

total = 0 inside a loop causes only the latest input to be kept. What is the fix?

A loop should visit indices 0 to 4. Which OCR loop fits?

Your score

Copy it from the score screen as soon as it appears.

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Key question

Can code run successfully and still give the wrong answer?

Review and improve

Use review mode. Explain one correction and why the correct answer fits. Implement the repaired countdown in Python. Confirm it prints 3, 2, 1 and stops. Change the initial value to 1 and check again.

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Trace $x = 3$, $y = x + 2$, $x = y * 4$, `print(x)`. Give y and the output. [2]

b) A loop starts $n = 1$ and repeats while $n < 4$ but never updates n . Explain and correct the fault. [3]

c) An algorithm reports B wins whenever A is not greater than B. Explain a required refinement. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Interpret algorithms using sample inputs.			
Correct faulty conditions and updates.			
Complete missing algorithm steps.			